

Executive Summary: House Price Prediction & Exploration

1 Data Exploration & Cleaning Observations

Initial analysis was conducted on a housing dataset containing structural and qualitative real estate metrics. The primary objective was identifying features driving property value and constructing a predictive baseline.

- **Dataset Scale:** The baseline data contains exactly **545 rows** and **13 columns**, with **price** assigned as the continuous target variable (y).
- **Data Integrity:** A structural null scan revealed **0 missing values** and **0 duplicate rows** across the entire dataset, confirming high initial data quality.
- **Categorical Encoding:** Seven qualitative features (**mainroad**, **guestroom**, **basement**, **hotwaterheating**, **airconditioning**, **prefarea**, and **furnishingstatus**) were converted into numeric formats using One-Hot Encoding (**drop_first=True**) to enable algorithmic training without introducing multicollinearity.

2 Model Performance & Evaluation

The processed records were divided into an 80% training and 20% testing split. A classical Linear Regression model was evaluated against an ensemble Random Forest Regressor using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

Table 1: Model Performance Evaluation Matrix

Evaluation Metric	Linear Regression	Random Forest Regressor
Mean Absolute Error (MAE)	\$970,043.40	\$1,021,546.04
Root Mean Squared Error (RMSE)	\$1,324,506.96	\$1,400,565.97
R^2 Score (Accuracy)	0.6529	0.6119

- **Model Superiority:** The **Linear Regression** model outperformed the **Random Forest Regressor**, successfully capturing **65.29%** of the variance in house prices compared to the ensemble's 61.19%.
- **Error Constraints:** The MAE shows that typical predictions deviate from the true property values by roughly \$970,043, which represents the standard residual spread of this target market.

3 Visualization Insights

Graphic analysis using Seaborn and Matplotlib revealed several distinct patterns within the data:

- **Price Skewness:** The target variable distribution is visibly right-skewed, indicating a market dominated by moderately priced residential units alongside a small cluster of high-value luxury estates.
- **Core Drivers:** Total plot `area` emerged as the strongest positive numeric driver of price, followed immediately by core utility metrics: the number of `bathrooms` and vertical `stories`.
- **Premium Variables:** Qualitative locational and comfort choices—specifically the presence of `airconditioning`, proximity to a `mainroad`, or being located in a preferred neighborhood (`prefarea`)—showed noticeably strong positive correlations with higher property values.